MUHAMMAD AMIRUL FAHMI BIN MOHD REDWAN

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SUMMARY

Fresh Computer Engineering graduate from Universiti Tun Hussein Onn Malaysia (UTHM) with hands-on experience across **software development, AI/ML, mobile applications, and embedded systems**. Proven track record of designing, building, and deploying real, working projects from concept to completion, including an **award-winning Android application** powered by machine learning, and an **IoT-integrated smart irrigation system**. Comfortable working across the **full development lifecycle**, from requirement gathering to deployment, with strong **problem-solving, analytical, and communication skills** developed through cross-functional academic projects and internships. Curious, adaptable, and eager to bring this technical foundation to a growing team.

CORE TECHNICAL SKILLS

Languages & Frontend: Python, JavaScript, TypeScript, Java, C++, Dart, React, Flutter, Three.js, GSAP, HTML, CSS.

AI & ML: YOLOv8/v9, scikit-learn, Random Forest, Decision Tree, Roboflow.

Data & Cloud: SQL, MySQL, Firebase, Supabase, Cloud Firestore.

Engineering: MATLAB, Verilog, Proteus, AutoCAD.

Networking: Cisco Packet Tracer, Wireshark, eNSP, PuTTY.

Tools: Git, GitHub, VS Code.

ACADEMIC QUALIFICATION

March 2023 – July 2026: Bachelor of Electronic (Computer) Engineering with Honours, University Tun Hussein Onn Malaysia (UTHM)

- Final Year Project: AI-Powered Android Application for a Real-Time Queue Management System using Random Forest Predictive Analytics in a Hospital Environment (Gold Medal, Final Year Project Industry Award).

July 2019 – July 2022: Diploma in Electronic Engineering (Communication), Politeknik Port Dickson

KEY PROJECT EXPERIENCE

Staff Calendar Management System (TalentBank)

- Built a backend calendar system for TalentBank staff with full event management functionality (create, edit, update, delete), removing the need for staff to touch any code to manage scheduling.
- Used AI-assisted coding tools throughout development to accelerate implementation, debugging, and iteration, allowing more focus on scheduling logic and user experience.

AI-Powered Mobile Queue Management System (Hospital Environment)

- Developed a mobile-based queue management system for the Android platform integrated with Random Forest Regression predictive analytics to forecast patient waiting times and optimise clinic workflows.
- Engineered a real-time monitoring feature to detect and mitigate hospital overcrowding via live queue status updates on a mobile interface.
- Built a cross-platform mobile application using Dart, and was awarded a Gold Medal in the Final Year Project Industry Award.

AI-Integrated Smart Irrigation System with Rainwater Harvesting

- Developed an intelligent irrigation prototype combining IoT monitoring and AI-driven predictive analytics, and secured 2nd Place in the Agriculture Category.
- Trained a Decision Tree model to predict weather patterns and rainfall probability, and deployed it for offline, on-device inference on a microcontroller, enabling real-time automated irrigation without reliance on cloud connectivity.
- Awarded a Gold Medal in the Integrated Design Project (IDP) Showcase Day.

Image Recognition Model via Dataset Augmentation & YOLO Predictive Analysis

- Developed a real-time object detection system integrating YOLOv8/YOLOv9 within a Cloud Computing environment.
- Curated and processed a dataset of 500+ images from Kaggle and Open Image Dataset V7, applying augmentation techniques (flipping, rotation, blurring) via Roboflow.
- Optimised model performance through comparative training analysis across a minimum of 100 epochs for high precision and low loss.

Workstation Safety Checklist Mobile Application

- Developed a digital mobile application using Flutter to modernise and streamline workplace safety inspections, replacing manual paper-based logs.
- Implemented Firebase Authentication with role-based access control for operators and supervisors.
- Integrated a Cloud Firestore backend for real-time data sync, enabling a live supervisor compliance dashboard.

PROFESSIONAL EXPERIENCES

July 2025 – Oct 2025: Internship as Maintenance, Media Prima Berhad

- Executed network switch resets using PuTTY and validated successful restoration of default settings across all input parameters.
- Delivered end-to-end technical support for major live productions (e.g. Anugerah Melodi, family singing shows), ensuring zero service disruptions and high-quality broadcast delivery.
- Checked network switch ports for connectivity/functionality and conducted internet speed testing to confirm stable bandwidth distribution.

March 2022 – August 2022: Internship as Maintenance, Media Prima Berhad

- Assisted the technical maintenance team in ensuring seamless broadcast operations, handling maintenance tasks and supporting live event setups.
- Successfully supported the live broadcast of Anugerah Juara Lagu and Hari Raya Aidilfitri shows with zero downtime.
- Diagnosed and fixed unstable or fluctuating signal issues across various departments.

REFERENCES

Available upon request.